THE LOWER SHRIEK OF A DERIVED TENSOR PRODUCT IS THE TENSOR PRODUCT OF THE LOWER SHRIEKS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the diagram of categories, the homotopy computation of the morphism sets, the application of the auxiliary lemma, and the final specialisation through the product-categories lemma.

Editorial note. The label lemma-eilenberg-zilber names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the six cited inputs are kept as external pointers with their canonical Stacks tags (the example-labelled pointer is recorded with kind section as the assembler requires).

Allowed standard prerequisites (editorial).

- The category-to-point situation and the functor L pi_!. Same chapter, the example describing the situation of a category over a point with its sheaves of rings and the functor L pi_!. Supplies the ambient setting and the functor whose compatibility with the tensor product is asserted.
- Computation of L pi_! by a cosimplicial resolution. Same chapter, the lemmas computing L pi_! through a cosimplicial object and the auxiliary computation for products. Supplies the identifications used for the diagonal and the product situation.
- Commutation of the pullback with the derived tensor product. Same chapter, the lemma stating that the pullback of a derived tensor product is the tensor product of the pullbacks. Supplies the identity g^{-1}K = K_1 tensor^L K_2.

Source reference. Original reference 08PF: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 10016–10016. Prerequisite pointer invoked at proof line(s) 10352; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08Q9: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 10229–10229. Prerequisite pointer invoked at proof line(s) 10354; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original reference 08QA: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 10298–10298. Prerequisite pointer invoked at proof line(s) 10382; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08QB: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 10316–10316. Prerequisite pointer invoked at proof line(s) 10389; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 079U: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3684–3684. Prerequisite pointer invoked at proof line(s) 10388; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08Q4: sources/stacks-chow/tags.txt, source lines 11047–11047. Prerequisite pointer invoked at proof line(s) 10381; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Notation and assumptions as in Example 08PF. If there exists a cosimplicial object $U_\bullet$ of $\mathcal{C}$ such that Lemma 08Q9 applies, then*

$$L\pi_!(K_1 \otimes^{\mathbf{L}}_{\underline{B}} K_2) = L\pi_!(K_1) \otimes^{\mathbf{L}}_{B} L\pi_!(K_2)$$

for all $K_i \in D(\underline{B})$.

Proof. Consider the diagram of categories and functors

$$\begin{array}{ccccc} & & & & \mathcal{C} \\ & & & \overset{u_1}{\nearrow} & \\ \mathcal{C} & \xrightarrow{u} & \mathcal{C} \times \mathcal{C} & & \\ & & & \underset{u_2}{\searrow} & \\ & & & & \mathcal{C} \end{array}$$

where u is the diagonal functor and u_i are the projection functors. This gives morphisms of ringed topoi g, g_1, g_2. For any object (U_1, U_2) of $\mathcal{C}$ we have

$$\mathrm{Mor}_{\mathcal{C}\times\mathcal{C}}(u(U_\bullet), (U_1, U_2)) = \mathrm{Mor}_{\mathcal{C}}(U_\bullet, U_1) \times \mathrm{Mor}_{\mathcal{C}}(U_\bullet, U_2)$$

which is homotopy equivalent to a point by Simplicial, Lemma 08Q4. Thus Lemma 08QA gives $L\pi_!(g^{-1}K) = L(\pi \times \pi)_!(K)$ for any K in $D(\mathcal{C} \times \mathcal{C}, B)$. Take $K = g_1^{-1}K_1 \otimes^{\mathbf{L}}_B g_2^{-1}K_2$. Then $g^{-1}K = K_1 \otimes^{\mathbf{L}}_{\underline{B}} K_2$ because $g^{-1} = g^* = Lg^*$ commutes with derived tensor product (Lemma 079U). To finish we apply Lemma 08QB. □